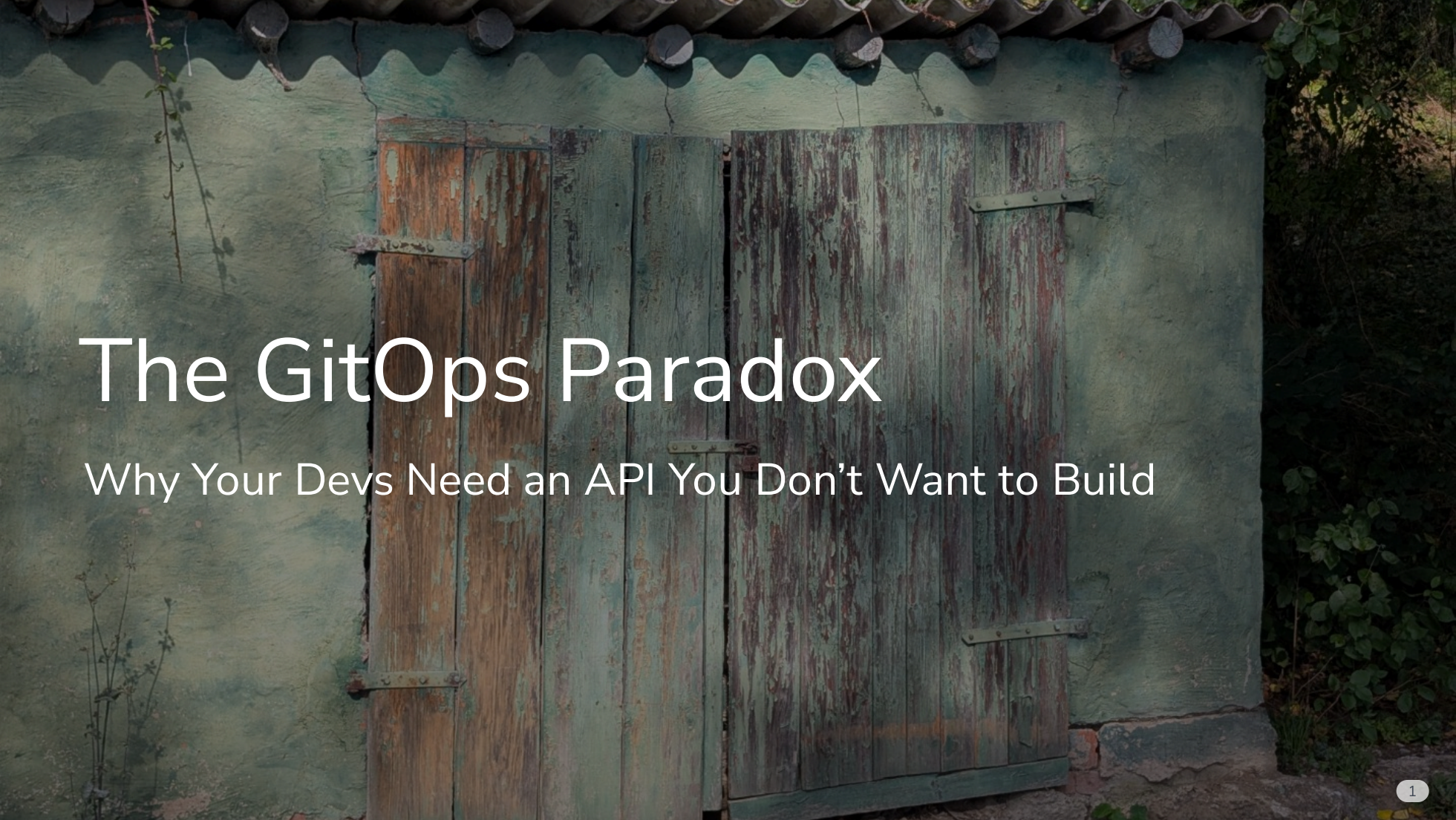


The background of the slide is a photograph of a weathered wooden door. The door is made of vertical planks and has a greenish-blue paint that is peeling and chipped away, revealing the natural wood underneath. It is secured with several metal hinges and a latch. The door is set into a wall with a similar textured, greenish-blue finish. Above the door, a portion of a corrugated metal roof is visible. To the right of the door, there is some dark green foliage. The overall lighting is somewhat dim, giving the image a moody, rustic feel.

The GitOps Paradox

Why Your Devs Need an API You Don't Want to Build



Who am I?

- Simon Koudijs
- ConfigButler

Experience

- Software engineer / DevOps engineer

Founder

- Quit my fulltime job to work on this fulltime
- Implementing this idear in the open
- ConfigButler



Will mine fly as
well?

I ♥ Files

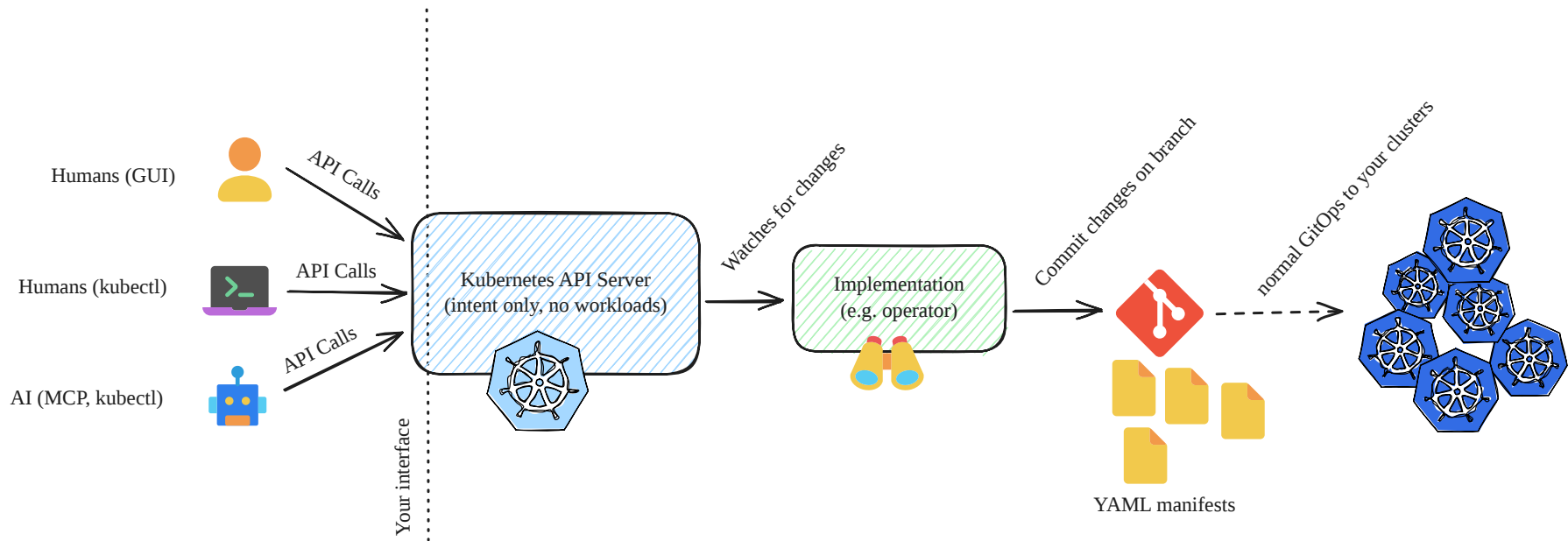
I ♥ Git

I ♥ GitOps

Pfff now I need to
create a PR with
the right YAML 😞

Why?

	GitOps	Reverse GitOps
Reproducible	✓	✓
Process	✓	✓
Full history	✓	✓
Rollbacks	✓	✓
API First 😊		✓
Fast type checking		✓



[main](#) 27 Branches 30 Tags

[Add file](#)
[Code](#)
 **github-actions[bot]** and **sunib** chore(main): release gitops-reverser 0.19.2 ✓ eb67c05 · last week 430 Commits

.devcontainer	chore: Get flux onboard to simplify the makefile and make it ...	last week
.github	chore: Fix linter and also run the bi-direction tests	last week
.vscode	fix: One edgecase where a secret is deleted (is now recreated)	2 weeks ago
api/v1alpha1	chore: refining age secret generation	last month
charts/gitops-reverser	chore(main): release gitops-reverser 0.19.2	last week
cmd	chore: First small step to HA	last week
config	chore: First small step to HA	last week
docs	docs: Improving docs	last week
hack	chore: Improving the test and keep forwards active	last week
internal	chore: Improving the test and keep forwards active	last week
test	chore: Lower wait times so that we can quickly validate it	last week
.dockerignore	feat: Use sops to encrypt secrets	last month
.gitignore	feat: Making big steps in Makefile understanding, but it's still ...	3 weeks ago
.golangci.yml	chore: Bump versions	last month
.release-please-manifest.json	chore(main): release gitops-reverser 0.19.2	last week
.yamllint.yml	chore: Refactoring away cyclomatic complexity	5 months ago
AGENTS.md	chore: e2e test for bi-directionality	last week
CHANGELOG.md	chore(main): release gitops-reverser 0.19.2	last week
CONTRIBUTING.md	docs: More info on deploying this thing	6 months ago
Dockerfile	chore: Get flux onboard to simplify the makefile and make it ...	last week

About



Reconciles Kubernetes API resources into a git destination as clean deployable YAML files (audit trail through git commits).

[audit-log](#) [gitops](#) [config-as-data](#)
[kubernetes-resource-model](#)
[reverse-gitops](#)
[Readme](#)
[Apache-2.0 license](#)
[Contributing](#)
[Activity](#)
[Custom properties](#)
[3 stars](#)
[0 watching](#)
[0 forks](#)
[Audit log](#)
[Report repository](#)

Releases

30

 **gitops-reverser: v0.19.2** Latest
last week

[+ 29 releases](#)

Packages

4

 **gitops-reverser-ci**
 **gitops-reverser**
 **gitops-reverser-devcontainer**
 **charts/gitops-reverser**

Contributors

3

Voting demo

- A live voting app is always nice
- This one is truly cloud native
- The presentation as well



Someone votes

The API receives a

`QuizSubmission`.

The synced repo turns that into a normal file.

```
apiVersion: examples.configbutler.ai/v1alpha1
kind: QuizSubmission
metadata:
  name: example
  namespace: vote
spec:
  answers:
    - questionId: q1
      singleChoice: Software engineer
    - questionId: q2
      singleChoice: Git
    - questionId: q3
      singleChoice: No
    - questionId: q4
      singleChoice: Curious
  sessionRef:
    kind: QuizSession
    name: demo
  submittedAt: "2026-02-05T12:34:00Z"
```

```
$ kubectl get quizsessions -n vote
NAME      STATE  TITLE
demo      live   KubeCon 2026 Demo
$ kubectl get quizsubmissions -n vote --watch
NAME      SESSION  AGE
example    demo      2m
$ git log --oneline --decorate --all -5
* 65b3c71 [CREATE] examples.configbutler.ai/v1alpha1/quizsubmissions/example
* e518c65 [CREATE] examples.configbutler.ai/v1alpha1/quizsessions/feedback
* fb999fd [CREATE] examples.configbutler.ai/v1alpha1/quizsessions/demo
```

Rendered output - on stage this was a live terminal

Demo takeaway

Not workloads. Just editable intent objects in Git.

Plain object

QuizSession

Plain object

QuizSubmission

Editable

with YAML

Trackable

as files in Git

```
demo-repo/  
adam/rai/8f/  
  examples.configbutler.ai/  
    v1alpha1/  
      quizsessions/vote/  
        demo.yaml  
        feedback.yaml  
      quizsubmissions/vote/  
        example.yaml
```

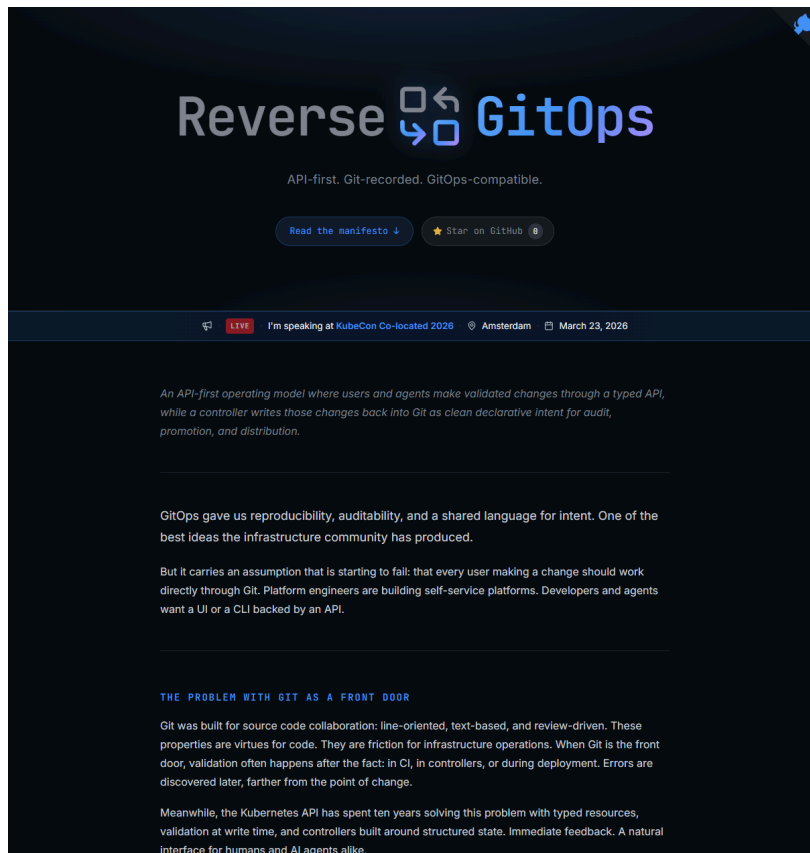
PDF snapshot - on stage this was studied by the APL

The paradox

- We shield people from the Kubernetes API.
- We force them to use Git as the write path.
- Then we build custom APIs and portals anyway.
- So why not use Kubernetes itself as the typed API for intent?
- Keep GitOps. Change the front door.



reversegitops.dev



The image is a screenshot of the 'Reverse GitOps' website. The header features the title 'Reverse GitOps' in a large, light blue font, with a logo consisting of three blue squares and arrows. Below the title is the tagline 'API-first. Git-recorded. GitOps-compatible.' and two buttons: 'Read the manifesto ↓' and '★ Star on Github'. A dark blue horizontal bar contains a 'LIVE' badge and the text 'I'm speaking at KubeCon Co-located 2026', 'Amsterdam', and 'March 23, 2026'. The main content area has a dark background with white text. It starts with a paragraph: 'An API-first operating model where users and agents make validated changes through a typed API, while a controller writes those changes back into Git as clean declarative intent for audit, promotion, and distribution.' This is followed by a section titled 'GitOps gave us reproducibility, auditability, and a shared language for intent. One of the best ideas the infrastructure community has produced.' and another paragraph: 'But it carries an assumption that is starting to fail: that every user making a change should work directly through Git. Platform engineers are building self-service platforms. Developers and agents want a UI or a CLI backed by an API.' Below this is a section titled 'THE PROBLEM WITH GIT AS A FRONT DOOR' in all caps. The text continues: 'Git was built for source code collaboration: line-oriented, text-based, and review-driven. These properties are virtues for code. They are friction for infrastructure operations. When Git is the front door, validation often happens after the fact: in CI, in controllers, or during deployment. Errors are discovered later, farther from the point of change.' The final paragraph states: 'Meanwhile, the Kubernetes API has spent ten years solving this problem with typed resources, validation at write time, and controllers built around structured state. Immediate feedback. A natural interface for humans and AI agents alike.'

Reverse GitOps

API-first. Git-recorded. GitOps-compatible.

[Read the manifesto ↓](#) [★ Star on Github](#)

LIVE I'm speaking at KubeCon Co-located 2026 Amsterdam March 23, 2026

An API-first operating model where users and agents make validated changes through a typed API, while a controller writes those changes back into Git as clean declarative intent for audit, promotion, and distribution.

GitOps gave us reproducibility, auditability, and a shared language for intent. One of the best ideas the infrastructure community has produced.

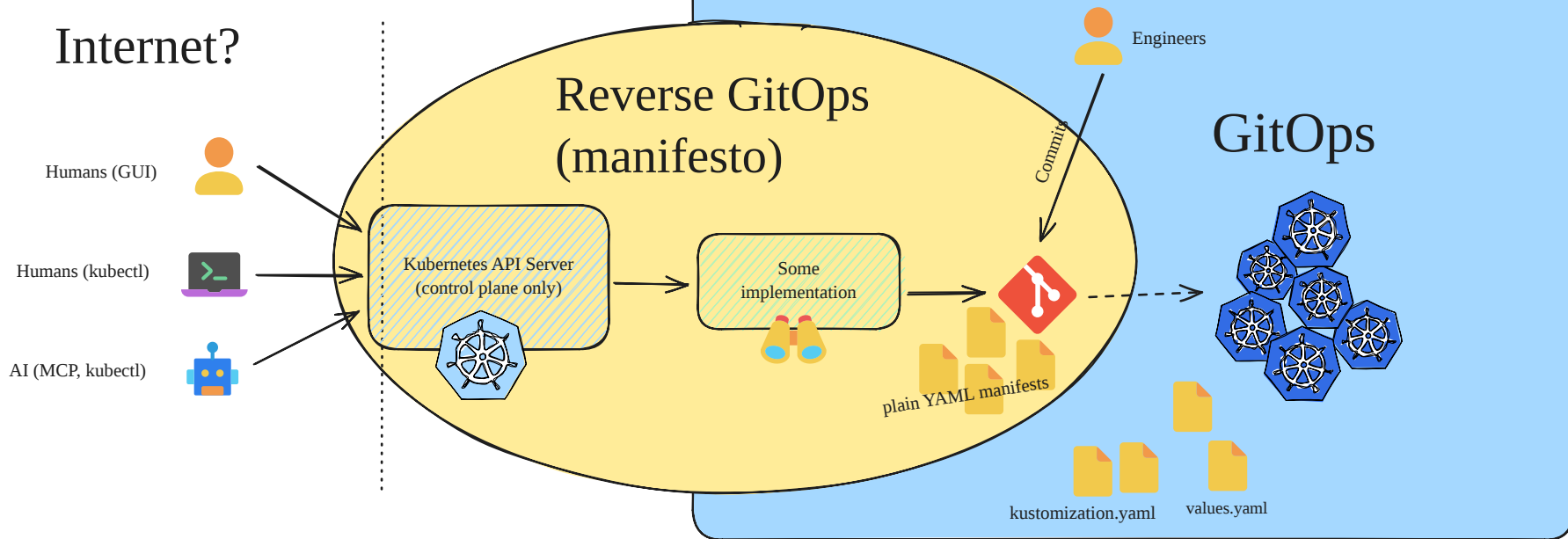
But it carries an assumption that is starting to fail: that every user making a change should work directly through Git. Platform engineers are building self-service platforms. Developers and agents want a UI or a CLI backed by an API.

THE PROBLEM WITH GIT AS A FRONT DOOR

Git was built for source code collaboration: line-oriented, text-based, and review-driven. These properties are virtues for code. They are friction for infrastructure operations. When Git is the front door, validation often happens after the fact: in CI, in controllers, or during deployment. Errors are discovered later, farther from the point of change.

Meanwhile, the Kubernetes API has spent ten years solving this problem with typed resources, validation at write time, and controllers built around structured state. Immediate feedback. A natural interface for humans and AI agents alike.

Internet?



1. API First

- Use Kubernetes as the front door
- Give humans, GUIs, and AI one typed write path
- Keep it separate from your workload clusters
- Reuse auth, authz, CRDs, status, and controllers



2. Capture Intent, Not Implementation

- This is not API-only
- Keep resources declarative and reversible
- Store clean intent, not generated implementation details
- Think CRDs, Crossplane, KRO, and similar patterns



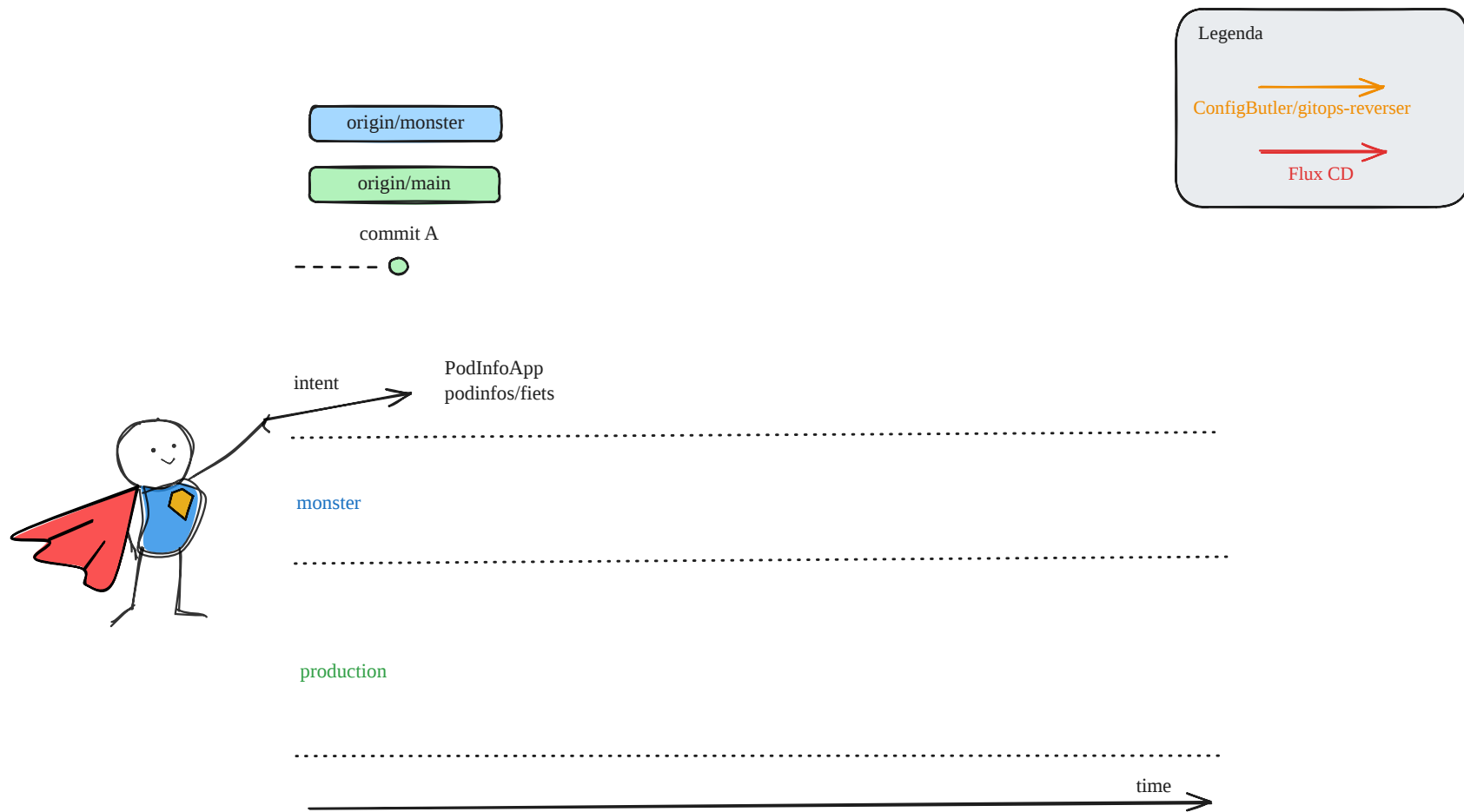
KRO as example

```
apiVersion: kro.run/v1alpha1
kind: PodInfoApp
metadata:
  name: fiets
  namespace: podinfos-intent
spec:
  replicas: 1
  message: The Dutch word for bicycle is "fiets". We have a lot of them. 🚲
  color: "#115711"
  logoURL: https://demo.configbutler.ai/podinfo-assets/podinfos-fiets.jpg
```

3. GitOps Applies

- Git still gives you review, history, rollback, and promotion
- The intent folder stays valuable and portable
- Existing GitOps tooling can keep doing the delivery
- The write path changes, not the deployment model






```
$ kubectl apply -f intent.yaml
```

```
podinfoapps.kro.run/fiets created
```

```
podinfoapps.kro.run/gevel created
```

```
podinfoapps.kro.run/paal created
```

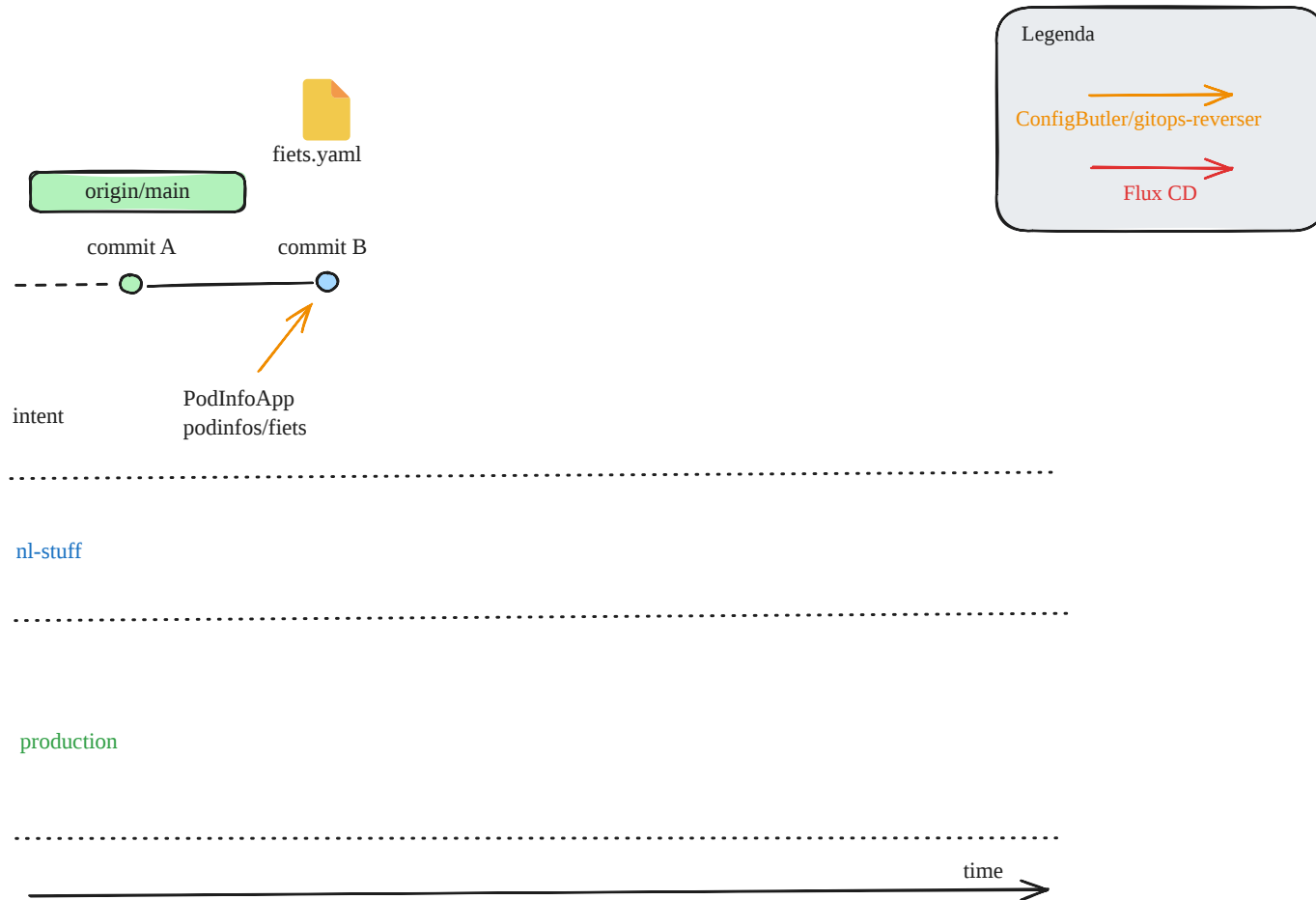
```
$ watch -n2 "git log --oneline --graph --decorate --all --color"
```

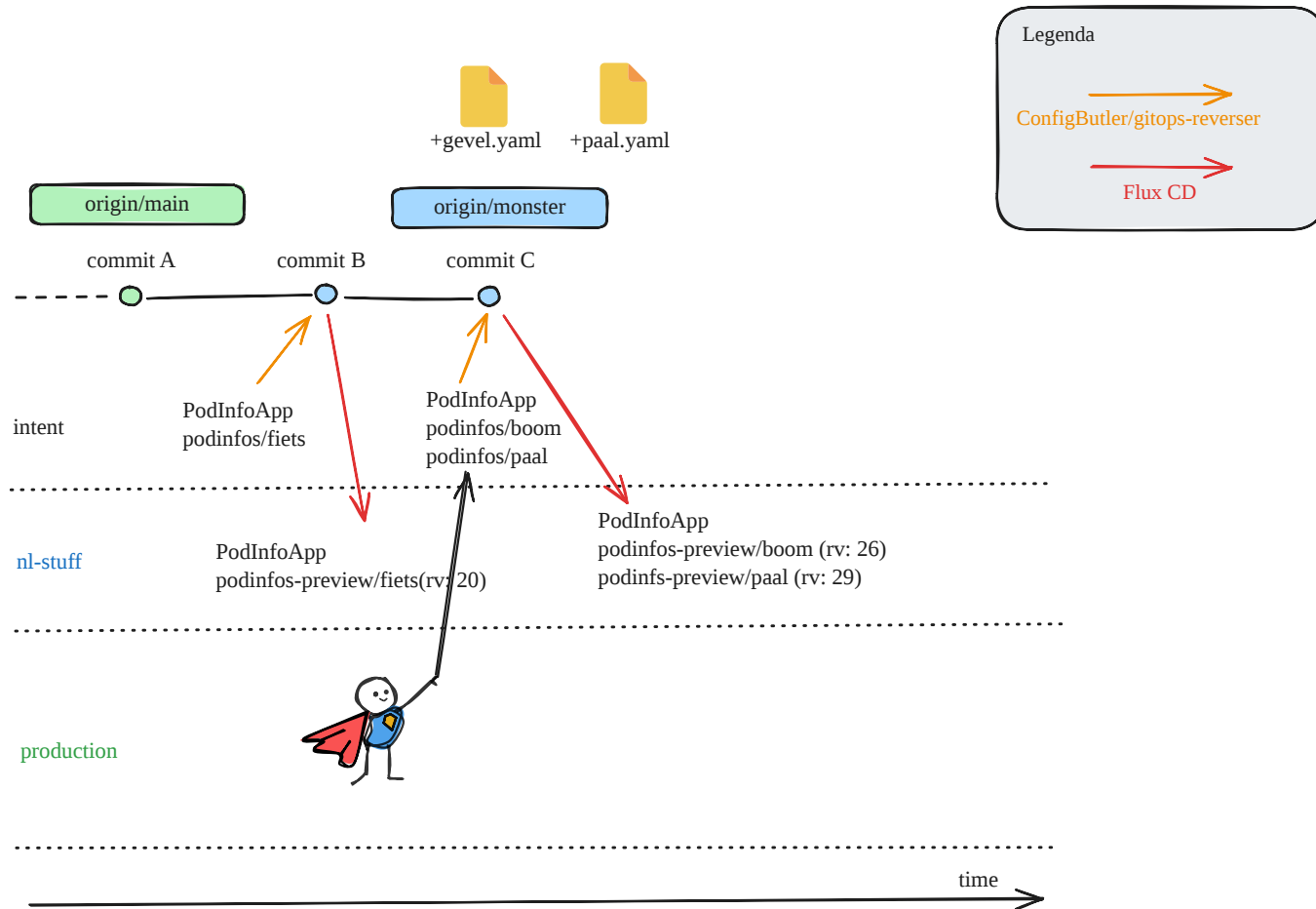
```
* 07a6eed (origin/holland-stuff) [CREATE] kro.run/v1alpha1/podinfoapps/paal
```

```
* 12f268e [CREATE] kro.run/v1alpha1/podinfoapps/gevel
```

```
* 2aeac44 [CREATE] kro.run/v1alpha1/podinfoapps/fiets
```

```
* 29e747e (HEAD -> main, origin/main) [UPDATE] v1/namespaces/vote
```





```
$ merge-holland-stuff.sh
```

```
0/5 Ensuring Gitea port-forward
```

```
1/5 Fetching origin
```

```
2/5 Checking out holland-stuff
```

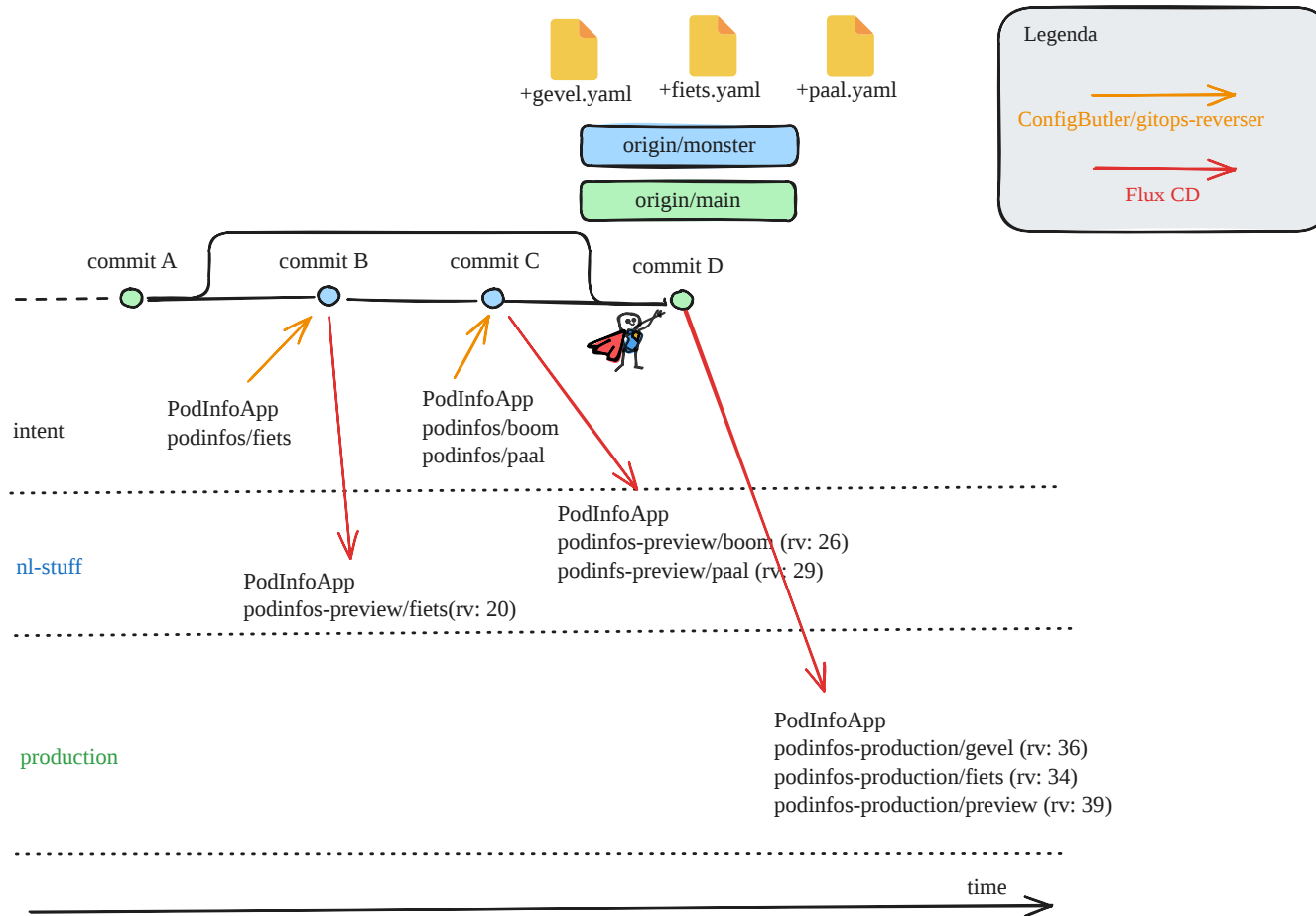
```
3/5 Checking out main
```

```
4/5 Merging holland-stuff into main
```

```
Done: holland-stuff merged into main and deleted locally and on origin
```

```
$ watch -n2 "kubectl get podinfoapps -A"
```

NAMESPACE	NAME	READY	AGE
podinfos-intent	fiets	True	42s
podinfos-intent	gevel	True	39s
podinfos-intent	paal	True	36s

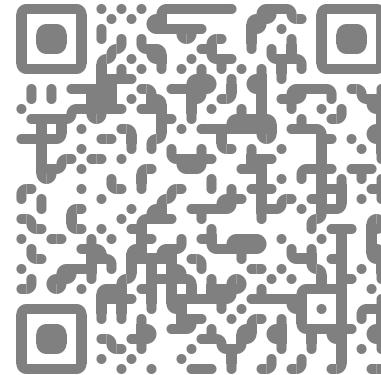


That's it

I would love your feedback.

This QR stays live for a while after the talk.

<https://calendar.app.google/FgYaoZog1dFpRG9Z7>



demo.configbutler.ai/feedback?code=null